

The human factor in the disaster resilience modeling of critical infrastructure systems

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ABSTRACT

Under major shocks such as natural or man-made disasters, the ability of human actors to maintain the safety, reliability and performance of critical infrastructure systems (CISs) is essential for improving the resilience of the systems. To understand human-system influence mechanisms and provide solutions for improving the disaster resilience of CISs, a growing number of studies are attempting to incorporate the human factor in the disaster resilience modeling of CISs. This paper presents a systematic review of such studies and discusses the state of the art on the disaster resilience modeling of CISs from a socio-technical systems perspective. The in-depth review of literature helped to identify and classify the main human factors affecting CISs resilience and also highlighted the social aspects of CISs resilience that are still overlooked in relevant studies. Moreover, several knowledge and technical gaps were identified and discussed while providing inferences to possible solutions or future research directions. The findings and discussions of this study could be of significant value to researchers and professionals in the area of CISs resilience that aim to identify and propose solutions to CISs resilience problems.

1. Introduction

Critical infrastructure systems (CISs) such as power systems, water systems, transportation systems, are essential for providing the basic goods and services that sustain modern societies and economies [1,2]. These systems are constantly under the threat of major shocks, as evidenced by the multiple natural and man-made disaster events that have hit the world in the past few decades, such as the Hurricane Sandy in eastern America, the Northeast blackout in the US and Canada, and the Zhengzhou floods in China. The need to improve the resilience of CISs intensifies as the dependency of societies on CISs is continuously growing. The resilience of CISs refers to a comprehensive measure of the disaster resistance, response, recovery and adaptive capacities of CISs [3]. Over the years, considerable engineering efforts have been made to improve the safety, reliability and resilience of CISs. However, CISs resilience challenges related to the social aspects of CISs are yet to be fully uncovered and addressed. CISs are complex socio-technical systems consisting of interdependent networks of technology and humans [4], that is, CISs still depend on the involvement and participation of multiple stakeholders including system managers, operators, policy makers, communities, and so on. These key stakeholders have the capacity to

control, intensify or mitigate the effects of disasters depending on their activities, performances, and behaviors at each disaster phase. The roles that humans play in CISs resilience find expression in their awareness of risks, understanding of current situations, efficacy of decisions or strategies, effectiveness of collaborations, timeliness and reliability of actions, and adaptability [5].

To improve the disaster resilience of socio-technical CISs it is important to clearly identify and understand the different human roles involved at each phase of disaster, and control the effects of the human factor on CISs resilience. The human factor describes human abilities, limitations, behaviors, reliability, and other human characteristics that are relevant to the design, operations, and management of engineering systems [6]. The human factor affects the disaster resilience of CISs in two main ways: (1) Humans are a major source of vulnerabilities to CISs. Humans are susceptible to making suboptimal decisions or even committing errors since they are neither perfect nor hyper-rational beings. From an analysis of major historical incidents in CISs, it was reported that the human factor contributed significantly to the occurrence and propagation of CIS disruptions. For example, human error, poor decision-making, and the delayed reaction of system operators were identified among the main causes of man-made disasters such as the

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Northeast blackout of 2003 in the United States [7] and the Italian blackout in the same year [8]. Statistical data shows that the forgetting behaviors of frontline workers in water supply systems cause at least 6.72% delays on planned schedules for system maintenance and restoration operations [9]. A number of other studies have demonstrated that human decisions, behaviors and actions may sometimes amplify the impact of disasters on CISs or negatively affect the post-disaster recovery of CISs [10–13]; (2) Humans are also the main capability for systems to respond and adapt to shocks and stresses. Timely engagement of the right stakeholders in planning activities such as pre-disaster infrastructure investments or collaborative actions for disaster relief is essential for promoting the robustness, response capacity and overall resilience of CISs [14]. The ability of key CIS stakeholders to collaborate, coordinate activities and resources, and align their preferences towards common goals has proven to have a significant impact on the level of resilience that CISs can achieve [2,14,15]. Similarly, the rapid changes in social habits or trends that accompany major disasters may also cause CISs to evolve or adapt by drastically altering some of the operational characteristics of the CISs [16–18]. For example, the recent COVID-19 pandemic brought major long-lasting changes in human communication and mobility trends due to social distancing and travel policies. The sudden surge in the reliance of people on internet services for activities that would have required displacements has considerably affected the way telecommunication and transportation systems are operated around the world [19]. Chester et al. [20] also argued that CISs are becoming obsolete in the face of new and unforeseen types of disasters due to outdated design principles that do not accurately consider the effects of social dynamics on the adaptive capacity of CISs. In summary, the impact that the human factor has on the resilience of CISs is significant and cannot be overlooked.

In order to identify critical influencing mechanisms affecting CISs resilience and develop efficient resilience management strategies, researchers have turned towards modeling and simulation techniques for analyzing CISs under complex disaster scenarios [21]. A wide range of advanced resilience modeling approaches, such as physics-based modeling, complex network modeling, agent-based modeling, system dynamics, mathematical modeling, and so on, have been proposed to capture the evolution of CISs in dynamic environments. However, a review of literature reveals that the majority of studies on the disaster resilience modeling of CISs are purely technocratic, with limited focus placed on the social aspects of CISs. Failure to include the effect of the human factor may potentially impact the accuracy and reliability of study results. To more accurately depict the complex nature of CISs, the systems should be modeled not just as networks of physical or cyber infrastructures but as socio-technical systems consisting of both technical and human aspects that dynamically interact. In recent years, a growing effort regarding incorporating the human factor in modeling the operations of different types of infrastructures has been observed. Several studies have emerged that investigate the impact of human reliability and decisions on the safety, reliability and resilience of infrastructures [10,11,22–25]. Such studies lay the foundation for understanding the role of the human factor in the resilience of infrastructures. However, from an infrastructure system perspective, similar studies are still in their early stages, emerging sporadically in different CIS domains. As a result, management problems and research questions about CISs resilience that can be explored using more advanced human factor-incorporated CISs resilience modeling techniques are yet to be unveiled.

Therefore, a comprehensive review of literature related to the human factor in the disaster resilience modeling of CISs is necessary to answer the following key questions: (1) What is the state of the art of this area of study? (2) What are the main human factors that affect the disaster resilience of CISs and need to be considered in resilience modeling? (3) What key social aspects of CISs resilience are still overlooked in existing studies dealing with CISs resilience modeling? (4) What are the main knowledge and technical gaps in this area of study, and possible future

research directions? In light of the above motivation, this paper aims to identify the human factors that affect the disaster resilience of CISs and review the methods used to consider these factors in CISs resilience modeling. This paper also aims to discuss current research trends, limitations, and possible future research directions. By achieving the above objectives, this study will contribute to advancing the understanding of the state of the art in this area, and highlighting the knowledge gaps that can be addressed by future research.

The remainder of this paper is organized as follows. Section 2 presents the methodology of the study, including details about the bibliographic search and review process. Section 3, discusses the identification, classification and modeling of human factors in CISs disaster resilience studies. Section 4 reviews literature on human factor consideration at different phases of CISs resilience. Section 5 discusses the research trends, limitations, and possible future research directions in this area of study. Lastly, Section 6 presents the concluding remarks.

2. Methodology

To identify the state of the art on human factor consideration in the disaster resilience modeling of CISs, a bibliographic search was conducted to collect and review all relevant literature. A systematic process was designed for this purpose which included four main steps: (1) definition of the scope of the review; (2) design of the bibliographic search; (3) screening of search results; and (4) review of selected literature.

(1) Definition of the study scope

Three criteria were adopted to define the scope of the present review:

- The review focused on studies related to sudden and short-lived CIS shocks including earthquakes, hurricanes, floods, and so on, and did not consider stresses such as climate change.
- Three phases of CISs resilience were considered, namely the pre-disaster planning phase, emergency response phase, and recovery and adaptation phase.
- The review focused on large-scale infrastructure systems, such as water supply systems, transportation systems, power systems, and so on, not individual infrastructures or facilities.

(1) Design of the bibliographic search

The bibliographic search was conducted in Web of Science (core collection) to collect articles published in journals and conferences indexed in the databases. The year of publication was not limited. Only articles (journal and conference) were considered for review. The databases were searched for articles containing keywords from all the following categories:

- Relevant concepts of CISs, including “infrastructure”, “system”, “network”, and “lifeline”, in the five considered infrastructure system domains, namely, power, water, transportation, gas, communication.
- Different phases of system resilience, including “planning”, “response” and “recovery”; and other related terms such as “preparedness”, “adaptation” and “resistance”.
- Relevant concepts of human factor, including “human factor(s)”, “ergonomic(s)”, “decision-making”, “human behavior”, and “human reliability”.

The above keywords were searched to be present in either the title, abstract, or list of keywords of relevant articles.

(1) Screening of search results

The initial search produced a total of 398 results. The title and abstract of all works identified during the search were checked for relevance. Papers that were found relevant were then read to examine their contents for a second round of screening. Specifically, the following criteria were used to discard inappropriate articles:

- Articles that did not describe any particular model, theory, or framework for considering the human factor in the disaster resilience modeling of CISs were discarded.
- Articles that focused on studying the human factor rather than its influence on the disaster resilience of CISs were discarded.

Following the screening process, 289 articles remained. An additional snowballing step was conducted to include important publications not indexed in the searched databases by identifying recurring citations in multiple selected articles. In the snowballing approach and after a last check for relevance, 31 additional articles were added to the sample, which led to a final sample of 320 articles. These articles formed the basis of this review, although occasionally hereinafter the authors also referred to publications outside of the review scope to provide additional contexts about certain concepts or techniques mentioned in the paper.

(1) Review of the selected literature

The selected articles were reviewed following two steps as follows. First, the articles were carefully reviewed to identify the human factors that affect the resilience of CISs, and the various approaches to incorporate those factors in CISs resilience modeling. Then, the selected articles were reviewed from the angle of individual resilience phases including pre-disaster planning, emergency response, and recovery and adaptation, to identify the progress in human factor consideration in the resilience modeling of CISs.

In the discussion section of this paper a bibliometric analysis is conducted to analyze publication trends in the subject area covered by this paper. The analysis presents the annual number of publications related to the human factor in the disaster resilience modeling of CISs by sector, normalized by the total number of publications on disaster resilience modeling in each sector. The scope of the bibliographic search to identify all the relevant publications on disaster resilience modeling in each sector is identical to the study scope presented in point “(1) Definition of the study scope” above, although the search terms and screening process do not include any human factor-related constraints.

3. Identification, classification and modeling of human factors in the disaster resilience of CISs

CISs are socio-technical systems that consist of closely interrelated human and technology aspects [4,26–28]. The Human Factors and Ergonomics Society (HFES) defines human factors studies as the study of human-system and human-human interactions, considering human abilities, behaviors, characteristics, limitations, and environment [6]. In the area of disaster resilience of CISs, human factors generally refer to those factors that affect the performance, reliability and behaviors of different classes of human actors involved throughout the lifecycle of disasters [29]. In this section, the human factors in the disaster resilience of CISs are identified and classified. Then, the existing methods for incorporating these factors in the resilience modeling of CISs are discussed.

3.1. Human factors in the disaster resilience of CISs and their classification

The human factors affecting the disaster resilience of CISs are specific to the human roles involved in the resilience management processes.

According to Trump et al. [30], there are three classes of stakeholders relevant to the resilience of CISs. The first group comprises the individuals tasked with the monitoring, control, maintenance or repair of the CISs. This group of frontline workers, referred to in this study as “system operators”, are responsible for making technical decisions and completing technical tasks that promote the capacity of CISs to absorb and recover from various shocks. The second group comprises high-level decision makers, such as system managers, policy-makers, risk mitigation agencies, and so on, who develop the disaster prevention, response and recovery strategies of CISs. This group of stakeholders, referred to in this study as “decision-makers”, guide infrastructure investments at different phases of disaster based on sectoral expertise and socio-economic-environmental constraints. The third group comprises individuals, communities, organizations or institutions that rely on the services provided by the CISs. This group of stakeholders is referred to in this study as “system users”. Although system users may not directly impact the operations of CISs, their public opinion, social characteristics (e.g., social equity, social capital) and so on, may significantly affect the resilience management policies and adaptation process of CISs. These three groups of stakeholders and their associated human factors are further discussed below in the context of the disaster resilience of CISs.

3.1.1. Human factors in CIS disaster resilience from the perspective of system operators

Due to their direct contact with the equipment and sub-systems of CISs, system operators play a crucial role in ensuring the safety, reliability and performance of CISs under different operational conditions (normal or disrupted). Their reliability and responsiveness determine the probability of occurrence of human-caused catastrophes, and may also play a significant role in controlling the propagation of disruptions and accelerating the recovery of CISs. For example, the 2003 Northeast blackout in the US happened as a result of several errors and poor responsiveness of system operators. Nevertheless, the rapid mobilization of recovery efforts ensured that power was restored within a relatively short period of time [31].

Numerous human factors may influence the ability of system operators, as individuals or groups, to form sufficient SA, make the right decisions, and successfully complete tasks. These factors increase failure risks and safety concerns in CISs and may also significantly affect the reliability and performance of these systems [32,33]. A number of studies have grouped and discussed the human factors relevant to system operators under two main classes namely, individual factors, and situational factors [34–36], as summarized in Table 1.

Individual factors refer to the factors specific to a particular individual, and can be grouped under three sub-classes, including their psychological state, physiological state, and capabilities. Psychological state refers to the mental condition of an operator during the exercise of his/her functions, such as high levels of stress, panic or impaired attention and perception, which can potentially lead to poor decision-making and human errors. Physiological state refers to the condition or state of an operator's body, such as fatigue from long tasks or injury sustained from a disaster event. Capabilities refer to the trained aptitudes of an operator such as knowledge and skills, or experience in similar disaster scenarios.

Situational factors (sometimes also known as task/system factors [34]) refer to operational characteristics or constraints of a CIS as they relate to a specific disaster scenario. Situational factors can be further broken down into several subclasses including system condition, tasks, procedures, working conditions, and team factors. System condition refers to the current state of a CIS before, during or following a disaster. Tasks refer to the characteristics, requirements or complexity of actions taken by system operators. Procedures refer to the availability and adequacy of work guidelines and communication protocols for different tasks and operational scenarios. Working conditions describe the environment in which operators have to work. Team factors refer to details about the coordination between multiple operators.

Table 1
Summary of human factors related to system operators.

Class	Subclass	Definition	Examples	Relevant publications
Individual factors	Psychological state	The mental or emotional condition of an operator during the exercise of his/her duties	Stress, impaired perception or appraisal	[37–43]
	Physiological state	The condition or state of bodily functions of an operator	Fatigue, pain, suddenness of onset and discomfort	[44]
Situational factors	Capabilities	The trained aptitudes of an operator	Knowledge, experience and skills	[38,40,45–48]
	System condition	The properties of a system that determine its performance levels	System capability, interface design, software reliability	[48–51]
	Tasks	The characteristics, requirements or complexity of actions	Interpretation of task, task complexity, frequency and repetitiveness of tasks	[11,13,35,52]
	Procedures	The format or type of work procedures	Logic structure, function / availability / validity of procedure	[2,5,11,16,23,24,29,47,52–61]
	Working conditions	The characteristics of the work environment	Physical access, lighting conditions, temperature, humidity, air quality, and noise	[49,62,63]
	Team factors	The characteristics of the collaboration among multiple operators	Team structure, team communication, team leadership	[24,56]

3.1.2. Human factors in CIS disaster resilience from the perspective of decision makers

CISs exist as part of a complex system-of-systems that integrate technology, governance, institutions, environment, society, and all other fundamental aspects of cities. CISs resilience heavily depends on the actions and interactions of the various individuals and groups that operate and use the systems. Thus, the resilience management of CISs is not an isolated matter but a common issue among all the major actors in society [15,17,29]. To promote the overall safety, wellbeing and sustainable development of modern cities, key stakeholders such as CISs managers, policy makers, leaders of various communities, institutions, and so on, need to collaborate and design strategies to preserve and enhance the resilience of CISs. These collaborations exist in various forms. First, as CISs become increasingly interdependent, managers of different CISs are required to jointly optimize the operations of the systems under both normal and disruption conditions to ensure proper functioning and performance of the CISs [64,65]. Second, infrastructure investments to strengthen the robustness and resistance capacity of CISs, such as investing in new infrastructures, updating safety standards and regulations, the pre-positioning of relief assets, resource allocation for CISs recovery, and so on, require negotiations and coordination between risk mitigation agencies, relevant service providers or companies, urban planning department, and other relevant institutions [66–68]. Third, policymaking for CISs needs to be mindful of social factors and drivers, such as public opinion, inflation, international relations, and so on, that may facilitate or hinder disaster response and recovery [16,69].

From the above discussion of the role and interactions of decision makers, it has been made clear in a number of prior studies that organizational factors are the main human factors that affect decision-making for the disaster resilience of CISs [70,71]. Organizational factors refer to both individual and group factors that determine how multiple stakeholders collaborate, negotiate and coordinate their

decisions or activities. These factors can be categorized under six subclasses, including goals and strategies, knowledge and heuristics, organizational structure, organizational resources, organizational management, and organizational culture. Table 2 summarizes and defines the different organizational factors that affect the disaster resilience of CISs.

3.1.3. Human factors in CIS disaster resilience from the perspective of system users

System users are key actors with a strong capacity to affect the performance and resilience of CISs. Infrastructures serve the needs of system users and should be able to respond and adapt to the dynamic changes in the demand for services and behaviors of system users before, during and after disasters. As the majority of studies on CISs resilience originate from engineering fields, a strong bias has been perpetuated in prior studies about the role of system users in CISs resilience, with system users considered only for their ability to alter the demand for services on CIS models [88,89]. For example, studies focusing on the resilience-based optimization of CISs networks generally abstract the system user role as a dynamic demand variable within a system model. Such models also do not clearly describe the mechanisms affecting the demand patterns of system users at different phases of disaster. A few other studies on the resilience of communities focus on modeling the prioritization of infrastructure services for different classes of system users. Such studies deduce the changes in demand for infrastructure services amidst disasters based on the vulnerability levels of different individuals or groups and other characteristics such as the role that a certain agent plays in the response or recovery process (e.g., hospitals, businesses, factories, and so on).

A broader perspective of the role and impact of system users on CISs resilience can be obtained by exploring studies from social science disciplines that are known for providing better insights into social factors,

Table 2
Summary of human factors related to decision makers.

Class	Subclass	Definition	Examples	Relevant publications
Organizational factors	Goals and strategies	The objectives and plan of action of individuals or groups	Preferences, interests; policies, guidelines	[33,43,49,68,72–78]
	Knowledge and heuristics	The decision criteria and methods of stakeholders	Biases, tools, intuition, logic	[79–83]
	Organizational structure	The relationship structures of stakeholders	Number of decision makers, control range, organizational levels	[14,45,67,68,82,83]
	Organizational resources	The material and non-material resources shared among stakeholders	Information resources, material resources, human resources, economic resources	[14,67,68,84,85]
	Organizational management	The systematic structure and processes of stakeholder collaboration	Leadership, coordination, control, management levels	[82–84,86]
	Organizational culture	The shared pattern of basic assumptions or standard practices among stakeholders	Safety culture, organizational climate, communication protocols	[83,87]

Table 3
Summary of human factors related to decision makers.

Class	Subclass	Definition	Examples	Relevant publications
Social factors	Service consumption	The service consumption behaviors of system users	Consumer / area profile, demand trends	[29,88,89,95–98]
	Social characteristics	The intrinsic characteristics of a community or society	Demographics, social stratification, social capital, social welfare	[69,91,92,99–101]
	Social dynamics	The relatively uncertain and fast changing attributes of society	Mobility trends, public opinion	[15,41,102,103]

dynamics and trends, as they relate to the built environment [18,30]. An important social factor discussed in recent literature is the rapid change of conditions in the Anthropocene that appears to exceed the adaptive capacity of CISOs [20,90]. Such conditions include emerging and disruptive technologies, inequity, ideological extremism, hyper-connectivity, and so on. Several decades ago, the majority of CISOs were not equipped to respond to disruptive human activities such as cyber-attacks because these were unforeseen risks based on the technological development of the time. As new technologies rapidly evolve, so do opportunities for new malicious activities that may catch CISOs off-guard. Additionally, the demand for new types of basic services may emerge over time as humans have been observed to rapidly adapt their activities and behaviors in response to disasters or other current living conditions. For example, it is likely that the surge in the reliance on internet services following the recent COVID-19 pandemic will affect the ability of cities to maintain and operate CISOs such as telecommunication and transportation systems at levels approaching previous conditions as designed and intended.

Another important human factor affecting the resilience of CISOs is social capital. The cohesion or bonds among system users, communities or institutions is a strong asset or resource for resilience [91]. Several studies have highlighted the critical role of social capital in the ability of communities to rebuild their infrastructures and other vital urban systems [69,92]. The availability of relief and recovery resources in the form of bonds, donations, volunteers, and so on, is determined by the quality and robustness of the social networks.

Other factors including social equity, public opinion, social responsibility, and so on, also strongly affect the design and implementation of policies that aim at improving the resilience of CISOs [93, 94]. At the same time, the dynamic behavior of CISOs under disasters may also significantly affect the resilience of society. For example, the cascading effects of disasters on CISOs may overtax the ability of societies to absorb adversity and sustainably develop [30]. It can therefore be seen that the resilience of CISOs and the resilience of society as a whole are interconnected. Table 3 summarizes and defines the different social factors that affect the disaster resilience of CISOs.

3.2. Approaches to incorporate human factors into CISOs resilience modeling

Traditional CISO models typically focus on modeling the topological and functional characteristics of a CISO, using empirical approaches, complex network (CN) modeling, agent-based modeling (ABM), system dynamics (SD), and so on. However, to model a socio-technical CISO, both the technological and human aspects of the CISO need to be considered. The latest socio-technical CISO models have incorporated aspects such as decision-making and operational tasks in CISO models. Following the review of the surveyed literature, the approaches that are commonly used to incorporate human factors in the resilience modeling of CISOs were identified and presented below.

(1) Optimization algorithms

Decision-making is critical in the resilience management of CISOs and mainly involves resource allocation, scheduling, or strategy selection,

which are generally formulated as optimization problems. In the surveyed literature, various methods to solve CISOs resilience optimization problems involving human factors as variables or constraints have been proposed. These methods commonly originate in operations research and network science [2,104]. For example, linear programming, multi-objective optimization, and mixed-integer programming, have been widely adopted for selecting disaster response and recovery actions, ranking decision alternatives, identifying critical decision influencing factors, and so on [67,105,106]. Other commonly used decision-making algorithms that aim to mimic human behavior or decision processes include analytic hierarchy process (AHP), genetic algorithms (GA), heuristic algorithms, Markov decision process (MDP), and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [107–111]. Among these, multi-attribute decision algorithms such as AHP and TOPSIS have been used to solve decision problems involving multiple (and sometimes conflicting) decision criteria, providing a good solution for investigating the correlative effect of different human factors on the resilience of CISOs.

(1) Markov's chain modeling

Markov chain is a widely adopted stochastic modeling process commonly used to mathematically model the transition from one functional or operational state, S_i , to another, S_j , of a system or system component [112,113]. State transition of system components is guided by a transition matrix or transition probability function that can be designed or adjusted to consider various external factors such as human factors and environmental constraints. Several studies have investigated the impact of human factors on CISOs by considering the reliability (human error probability) of system operators as a key variable of the transition probability function of the Markov model [5,114,115]. This approach can also be used to integrate any other human factors that can be represented as probability functions impacting the state of CISO components [116].

(1) Agent-based modeling (ABM)

ABM is a commonly used approach for modeling CISOs and human agents and simulating their decisions and actions. Agents can be modeled using different behavioral rules or principles to make them unique and capture heterogeneities and complex interaction dynamics among different agents. For the above reasons, ABM has been adopted in several studies involving decision-making agents and systems users with dynamic behaviors or patterns [11,60,117]. Another aspect for which ABM is well recognized is its capabilities in modeling and simulating complex socio-technical systems exhibiting emergent behavior [54,118, 119]. An emergence is an unpredictable or hardly predictable change in functional or operational patterns of a system that results from interventions or disruptions. For example, the recent COVID-19 caused people to change their travel behaviors, leading transportation systems to adapt to new mobility trends and demand. ABM can fairly easily capture such emergence thanks to its ability to represent important phenomena resulting from the characteristics and behaviors of individual agents and their interactions [118].

(1) Bayesian belief networks (BBNs)

BBNs are models that represent and quantify probabilistic relationships among variables. BBNs are sometimes used to model CISs and analyze various probabilistic processes such as failure propagation [120, 121]. The ability of BBNs to represent the relationship between complex influencing factors and handle large multi-source data has made it attractive for modeling human factors. The main application of BBNs for integrating human factors and CISs models is the estimation of the failure probability of CIS components considering uncertainties in human performance [56,122–124]. BBNs also provide a relatively simple way of analyzing the correlations among multiple human factors while also being able to perform simple impact assessments on CISs.

(1) System dynamics (SD) modeling

SD modeling is one of the main approaches used to model the influence mechanisms and impact of human behavior on systems in fields such as operations management, economics, and so on. SD modeling has been used in several CIS studies to identify various feedback loops and causal relationships in CIS operations involving human actors and decisions [125–127]. Growing interest in SD modeling among researchers in CISs resilience comes from the ability of this approach to simulate temporal behaviors of complex systems and assess the performance of alternative policy options [78]. SD models are able to predict the evolutionary behavior of CISs affected by key variables representing disaster-related policies, stakeholder preferences, public opinion and so on [78,128,129]. The high-level representation of system processes and interactions makes SD models relatively easier to develop and apply for real-life scenarios compared to models such as ABM models. However, SD models provide limited information about component-level dynamics and are therefore not adequate for supporting technical analyses and decisions [129].

(1) Other approaches

A few other approaches exist for incorporating human factors in CISs models that focus primarily on evaluating human related risks, and safety issues. They include: combining HRA techniques and Monte Carlo simulation process [10,12,32], fault tree analysis [130], fuzzy logic [131], input-output modeling [132,133], and discrete event simulation [134,135]. A few social science theories have also been adopted to model human decision-making behaviors in CIS. Some of these theories include game theory [68,106], prospect theory (PT) [136,137], and auction theory [67].

4. Human factor consideration at different phases of CISs resilience

The above review of the approaches for incorporating human factors in the resilience modeling of CISs lays the foundation for understanding how human-related problems are being addressed in CISs resilience studies. On this basis, this section reviews studies that have adopted the above approaches for incorporating human factors in CISs resilience modeling to address problems specific to each resilience phase of CISs. Specifically, this section explores the pre-disaster planning and preparedness, emergency response, and recovery phases of CISs resilience, focusing on the human roles involved at each stage.

4.1. Pre-disaster planning and preparedness in socio-technical CISs

Pre-disaster planning and preparedness is the first and perhaps the most important phase of CISs resilience. This phase involves the early warning and forecasting of disasters (including natural or man-made disasters, and major technical failures), predicting disaster impact, and selecting strategic measures to mitigate risks and minimize disaster

impact [138,139]. CIS managers and policy makers are key actors in this phase. System managers are responsible for identifying the critical components of their systems that require investments ahead of disaster and establishing preparedness action items to be reviewed by policy makers. The policymakers (such as local governments, disaster management agencies, and so on) are responsible for setting disaster management goals and objectives, identifying and evaluating risk mitigation strategies, allocating resources for undertaking action items, and reviewing/revising plans periodically. The ability of CISs decision-makers to identify risks and select optimal risk reduction strategies within a limited budget has a direct impact on the safety, robustness and performance of CISs when facing disasters. Below, works related to the main human roles involved in pre-disaster planning and preparedness of CISs, including CISs managers and policy makers, are reviewed in detail.

To support investment decision-making for risk reduction, CISs managers need to rank the criticality or importance of system components and identify the components that need to be prioritized for investment ahead of a disaster. In the majority of CISs studies, critical system components are identified and ranked based on various component importance measures (CIMs) [140], including topological-based CIMs such as average path length or network efficiency, and flow-based CIMs such as flow vulnerability and serviceability. However, due to the highly uncertain or risky nature of the decision-making environment, it is challenging to forecast disaster events and accurately predict the impact of a disaster on CISs. Therefore, there exists a risk of making wrong investment decisions that could result in significant losses or penalties. In addition, limited budgets may increase the challenges in deciding where to invest, and whether or not to invest in strengthening CISs ahead of disasters [138,141]. Psychological research provided a basis to suggest that when making decisions under uncertainty and risks, human factors under the behavioral regularities category, including decision heuristics or biases and loss aversion, may significantly affect human decisions [142]. Heuristics and biases are general decision-making strategies or mental shortcuts that reduce the cognitive burden associated with decision-making. Tversky and Kahneman [143] noted that people's mental shortcut to reasoning, when faced with uncertainties and risks, causes them to fall prey to irrational tendencies. A few studies adopted heuristic algorithms, GA, and MDP [107–111], to model and study different decision heuristics and biases in CIS planning. On the other hand, loss aversion proposes that decision-makers are not neutral to the outcome of their decisions. That is, different decision makers associate different values to losses and gains [144]. The attitude of decision-makers towards losses and gains can significantly affect their willingness to take certain risks and make certain investment decisions. For example, Masini and Menichetti [145] demonstrated that loss aversion can influence the decision of system managers to update their system components due to the uncertainties related to the willingness of end-users to pay higher service prices. Although limited research has explored the implications of decision-makers' loss aversion, a few studies have demonstrated that the prospect theory (PT) could potentially provide a solution for incorporating loss aversion in pre-disaster planning models [141,146]. For example, Feng and Xu [146] applied PT to analyze the risks and profits of capacity-over-offer behavior of CIS managers during bidding in a capacity market. PT helped define a penalty mechanism that could make certain decisions less profitable and riskier, hence less appealing to the decision-makers.

Concerning the policy maker role, numerous studies have developed analysis methods to inform pre-disaster planning and preparedness in CISs. Some prior studies have focused on developing metrics for measuring CISs resilience to support risk-informed policy development and strategy evaluation [147,148]. Other research has focused on testing different approaches to model the resource allocation or strategy selection processes, and evaluating the performance of CISs under different risk reduction strategies [149–151]. This section mainly

discusses the latter category of studies, particularly those in which the human factor was considered in the developed models. Some of the commonly adopted approaches in prior studies to model resource allocation or pre-disaster investment strategy selection include optimization, multi-objective decision-making, input-output modeling, and economic theory. Human factors are considered in models developed in prior studies in the form of decision criteria or constraints, and may include individual or organizational factors such as individual preferences, goals and policies, quality and capabilities of the decision-makers, and so on. For example, Pham and Hwang [149] proposed a multi-objective planning model for resource allocation that considers fairness as a decision criterion to improve the energy efficiency of multiple subsystems of a communication network. In addition, the selection of risk mitigation strategies may sometimes involve strategic games between multiple CISs stakeholders under a semi-decentralized decision-making environment in which the policy makers make the final decisions [138]. Therefore, several studies have adopted the game theory, specifically non-cooperative games, to study the interactions between the CISs stakeholders and investigate the influence of divergent goals and preferences on the effectiveness of pre-disaster planning strategies [106,152].

4.2. Emergency response in socio-technical CISs

Although the planning activities of CIS decision makers can help mitigate risks and reduce the impact of disasters, the systems may still suffer significant disruptions when subjected to adverse events. Following the direct impact of a disaster, a prompt and effective response is necessary to prevent, control or stop the propagation of failures in CISs. Otherwise, system failures may rapidly propagate in the CISs, potentially leading to global failure of the systems such as the case in the Northeast blackout of 2003 [31,96]. System managers and operators play key roles in ensuring the effectiveness of response decisions and operations as they need to react promptly with limited time and information to control the propagation of failure in the systems and minimize losses. System managers are involved in decision-making tasks such as response crew scheduling, crew assignment for emergency operations, repair vehicle routing, and so on. They are susceptible to relying on experience-based heuristics or intuition for decision-making under emergency conditions which may result in them making sub-optimal decisions [153–155]. System operators, on the other hand, are involved in monitoring system parameters and executing the emergency action items defined by the system managers. Their actions are error-prone since human reliability is significantly affected under emergency conditions due to a number of individual and task-related factors such as stress, unfamiliarity to event conditions, the complexity of emergency tasks, and so on. Poor decision-making and human error are among the main threats to the safety and reliability of engineering systems and growing effort is being made to understand and minimize their impacts [5,156,157]. Below, works related to the decision maker role in emergency response are first reviewed, followed by a review of works related to the CIS operator role.

Several methods, models, or algorithms have been proposed in prior studies for optimizing the allocation and deployment of emergency resources in interdependent CISs [134,135,158,159]. Although similar approaches are sometimes used in literature to develop resource allocation models for both pre-disaster planning and emergency response, resource allocation in the context of emergency response is very different from pre-disaster investments in terms of the nature of resources allocated and the decision-making environment. Immediately following a disaster, decision-makers might not have complete information about the actual state and functionality levels of their systems. This is due to factors such as inadequacy of communication or information sharing protocols, failure of control systems, delays or error in information sharing between different management levels, and so on. As a result, decision-makers often rely on a host of decision heuristics or

intuition for convenience and speed when information about the disaster-impacted system is limited [160]. Such heuristics include representative heuristic and availability heuristic. Representative heuristic refers to the tendency to judge the nature or likelihood of events by the extent to which they resemble a “typical” case, while availability heuristic refers to the tendency to judge the nature or likelihood of events based on available relevant experience or memories. Human decision heuristics favor speed and convenience over certainty or optimality of the decisions. To mimic and analyze the effects of such decision-making behaviors, a few studies have developed heuristics-based decision models for emergency response in CISs. These models aim to find an approximate solution to a problem more quickly when classic optimization methods are too slow. For example, Fritzen et al. [160] proposed an integrated use between constructive heuristic methods and integer programming by establishing a methodology for its completion. Weintraub et al. [158] developed a composite heuristic based on the “cluster first, route second” approach for a crew assignment problem in a power supply system.

Concerning the modeling of emergency response operations, a considerable amount of effort has been made in developing and testing advanced methods, such as HRA techniques, BN, FTA, bow-tie approach, and so on, for modeling and assessing the performance and reliability of operators in emergency tasks [24,161]. Fundamental studies in this area have attempted to unveil the combined effects of hazard conditions and human performance on the management of risks and safety of CISs [162, 163]. More recent studies have adopted methods such as Infrastructure Resilience-Oriented Modelling Language (IRML) and Markov’s chain to model and analyze the dynamic time-dependent resilience of CISs, and provided recommendations to improve the disaster response strategies of CISs [77,164]. During emergency response operations such as switching operations, dispatch operations, or emergency repair operations, individual SA, effective crew collaboration, and information sharing are among the major determinants of the robustness of CISs [56]. In the surveyed literature, two approaches have been adopted to model and assess the impact of the above factors on CISs emergency response operation, including the reductionist approach and first-principle approach [165]. The reductionist approach is a top-down approach that organizes CIS operations into human-system task sequences. It predicts human performance through an extended task and system analysis which is followed by an impact assessment on the CIS. Several studies have adopted this approach to develop models that can predict human performance in control rooms under emergency scenarios and the impact of human error on cascading failures in CIS networks [61,166]. The first-principle approach, on the other hand, is based on the mechanisms that underlie and cause human behavior [165]. It can model basic human cognition and integrate other domain knowledge such as experience, rules, learning, and so on. An example of a framework developed based on the first-principle approach is the adaptive control of thought-rational (ACT-R) [167]. Most HRA techniques such as Technique for Human Error Rate Prediction (THERP) [168], Cognitive Reliability and Error Analysis Method (CREAM) [11], Human Error Assessment and Reduction Technique (HEART) [169], and A Technique for Human Error Analysis (ATHEANA) [170], can be used in both reductionist and first principle approaches. Consequently, these techniques are more widely adopted for modeling and addressing problems related to human performance and reliability in CISs.

4.3. Post-disaster recovery of socio-technical CISs

Following the short-term disaster response processes of CISs, adequate long-term system restoration strategies must be developed to speed up the recovery of CISs, and improve their resilience [171]. The recovery process is usually a multi-agent and multi-system process that may involve interactions between CIS stakeholders from different CISs domains. Following major disaster events such as floods, hurricanes, earthquakes, and so on, multiple CISs may simultaneously suffer severe

damages that can take a few days, weeks, or even months to fully recover [7]. The resources involved in the restoration of the CISs under such conditions may easily exceed the budget of each CIS. Hence, additional resources would need to be provided by external sponsors or policy makers such as the local government to facilitate and accelerate the recovery process of CISs [138]. Due to the limited resource budget that can be allocated to each CIS, the competition for resources is intensified, making the selection of adequate restoration strategies considerably more challenging.

Several studies have attempted to solve the problem of optimal resource allocation and scheduling in the post-disaster recovery of competing CISs, primarily using optimization approaches. For example, Baidya and Sun [172] developed a LP optimization model for coordinating system restoration activities between interconnected networks. Mackenzie and Barker [173] proposed and compared two optimization-based models for determining the optimal resource allocation scheme to minimize the effect of direct and indirect impacts of disaster on competing interdependent CISs. Human-related constraints such as the basic service requirements of end-users, management objectives, economic policies, and so on, play a major role in the design of the optimization models for post-disaster modeling. Among the literature on optimization, some studies attempted to replicate more complex organizational behavior and recovery strategy selection processes of decision makers such as the dynamic adjustment of recovery goals and preferences based on current social needs [64,105]. The findings from the above studies revealed that by dynamically adjusting recovery decisions with respect to other socio-economic variables, decision makers can considerably improve the disaster resilience of CISs.

To model the dynamic interactions between CIS stakeholders involved in recovery planning, prior studies developed game theoretical models. Game theory is the most widely adopted method to study and understand decision-making involving multiple stakeholders [174]. It can describe the attitude, beliefs, and responses of decision-makers and the associated impact on the outcomes of decisions. Hence, game theory provides a good solution for modeling strategy selection for the joint-recovery of CISs [67,172]. Two main types of restoration games were proposed in the surveyed literature, namely cooperative games and non-cooperative games. In cooperative recovery games, all the players in the restoration game pursue the same goal. The players are not driven by self-interest but rather contribute to achieving a common goal. For example, Mohebbi et al. [68] adopted cooperative games to model the selection of restoration schedules for the joint-recovery of two interdependent water and transportation networks, and Coles and Zhuang [152] adopted cooperative games between relief organizations to model the allocation of the relief effort, both assuming the full cooperation, collective goals, and perfect beliefs of the players. However, in reality, the players involved in restoration games are potentially self-interested, which is the basis of non-cooperative games. In non-cooperative games, players make strategic moves aimed at maximizing their interests. For example, Sharkey et al. [175] modeled decision-making for interdependent network recovery based on non-cooperative games, in which the recovery strategy is revised at each iteration until a Nash equilibrium is achieved. Nash equilibrium is a state in which no further trade-offs can be made by any player. Smith et al. [106] proposed a decision-making model based on non-cooperative games that could consider the limited communication between players and its impact on the effectiveness of joint-recovery strategies of infrastructure networks. Due to the decentralized decision-making environment of joint-recovery operations, stakeholders have limited information exchanged between them [67]. Under limited information exchange conditions, a player's beliefs would deviate from the actual decisions being made by other players in a game, resulting in the selection of suboptimal strategies [176]. To provide a solution to this problem, Talebayan and Dueñas-Osorio [67] introduced an approach for modeling decentralized decision-making during joint-recovery of CISs that takes into account imperfect communication between decision-makers. This technique, known as judgement call,

aims to minimize communication delays and improve the speed of recovery strategy selection of interdependent CISs.

Although system users do not make decisions that directly affect the disaster recovery of CISs, they facilitate or impede crucial aspects of CISs recovery including information sharing, participation and collaboration among groups or institutions [29]. Aldrich and Meyer [92] proposed a framework describing the social factors among communities and their influence on the disaster recovery of the built environment. These factors, summarized as bonding, bridging and linking, are important variables of models studying the social capital of stakeholders in disaster recovery [69]. Combining system features and community features is important when modeling the recovery of CISs as services and resources are generally prepositioned or prioritized for the most vulnerable or high-profile communities [177]. Several studies on community resilience have demonstrated using techniques such as ABM and SD that the performance and behavior of infrastructures facilities during recovery scenarios vary in response to the dynamics of community needs and preferences [55,72,178]. Although not many analytical models or techniques to quantify the impact of social factors on the adaptive capacity of CISs exist, it is unanimously agreed among researchers from various fields that system users (their activities, needs and behaviors) are fundamental drivers of the adaptation of CISs to future shocks and stresses [20,29]. Hence, system users can influence the resilience of infrastructures and should be modeled as those with the capacity to transform the system performance rather than simple demand variables of infrastructure services.

5. Discussions

The above review has painted a clear picture of the state of the art around the human factor in the resilience modeling of CISs and set the foundation for discussing the research trends, knowledge and technical gaps, as well as possible future research directions. Each of these items is discussed in-depth in the following subsections.

5.1. Bibliometric analysis

Fig. 1 shows the annual number of publications by sector between 2000 and 2022 related to the human factor in the disaster resilience modeling of CISs, while Fig. 2 presents the percentage of human factor related publications between 2000 and 2022 normalized by the total number of studies on CISs disaster resilience modeling in each sector. It can be seen from Fig. 1 that in the last decade, the annual number of publications dealing with modeling the human aspects of CISs resilience experienced the fastest growth in the power, transportation and water

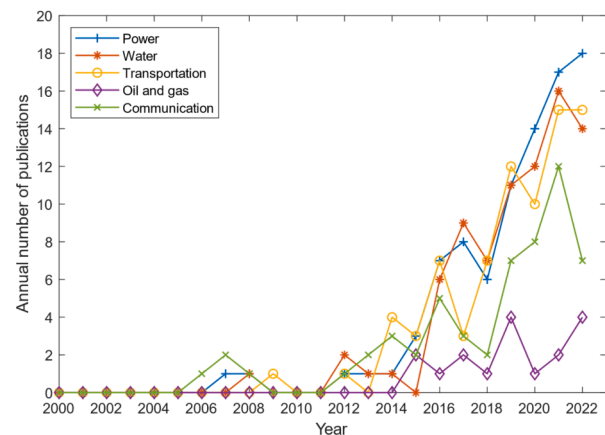


Fig. 1. Publication trend of literature related to the human factor in the disaster resilience modeling of CISs published between 2000 and 2022 by sector.

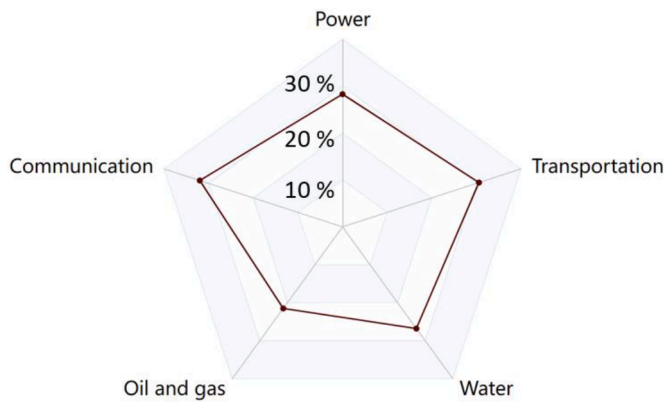


Fig. 2. Percentage of studies on CIs disaster resilience modeling within each CIs domain dealing with the human factor.

sectors. However, it can be observed that the percentages of resilience modeling studies from each sector that considered human factors do not differ considerably from one sector to another, except for the gas sector which seems to be lagging. At the time of this review of literature, the communication sector appears to be leading in terms of the percentage of CIs disaster resilience studies dealing with the human factor. The reason behind this observation could be that major disasters considerably affect human-human and human-system interactions that rely primarily on relatively vulnerable communication networks [179]. Communication systems are crucial for building situational awareness of the state of systems and society following major disaster events. They also play a central role in the information sharing among CIs stakeholders, and the coordination of emergency and recovery activities of CIs. Improving the performance and resilience of communication systems has therefore become a crucial matter in CIs related research. It was also observed from the literature review that the CIs domains with the largest variety of approaches and tools to model human behavior and performance in CIs resilience included the transportation and power system domains. For example, most human reliability analysis techniques used across different CIs domains were originally developed for power control rooms and aviation systems [5,11]. In summary, studies on the human factor in the disaster resilience modeling of CIs have seen steady growth in the various CIs domains. Publications are expected to keep growing as more and more attention is paid to understanding and managing the social aspect of CIs resilience.

5.2. Disaster resilience modeling of CIs considering multi-dimensional interdependencies

CIs are becoming increasingly interconnected and dependent on each other for their proper operation. Any disruption in one CIs may trigger ripple effects across multiple CIs that are difficult to predict, mitigate or eliminate [10,31]. In prior research, the effects of physical, cyber and geographical interdependencies on CIs disaster resilience have been extensively modeled and studied, revealing important influence mechanisms between physical components of CIs [21,96,180,181]. However, the logical interdependencies between CIs, which encapsulate the wider and more complex socio-economic and management interplays between CIs, have not been thoroughly studied and explained in prior studies, presenting a major knowledge gap in the reviewed literature. Rinaldi et al. [1] defines logical interdependencies of CIs as a control schema that links agents in one CIs to agents in another CIs without any direct physical, cyber, or geographic connection. Logical interdependencies between CIs mainly relate to management aspects of resilience, such as collaborative decision-making, information sharing, competition for resources, and conflicts of interest among CIs stakeholders. The review of literature reveals that limited is

known about the above management aspects of interdependent socio-technical CIs. Consequently, it is challenging to calibrate existing CIs resilience models or make reasonable assumptions about organizational behaviors [14,182]. Moreover, there is a lack of studies that have quantitatively analyzed the effects of policy regulations, multi-stakeholder negotiations, conflict resolution and resilience strategy coordination across interdependent CIs. Thus, important influence mechanisms that are crucial for the disaster resilience management of CIs are yet to be fully understood.

The above knowledge is crucial for developing CIs disaster resilience models that can accurately capture and simulate the interdependencies among socio-technical CIs. A few studies have attempted to assess the effects of logical interdependencies on the joint-recovery of interdependent CIs or pre-disaster infrastructure investments [67,68,85]. However, the models developed in those studies simply aim to optimize task scheduling and resource allocation problems between interdependent CIs using mostly technical criteria and constraints. The criteria and mechanisms governing the actual collaboration and coordination between the decision makers of interdependent CIs, such as their cooperative attitudes, personal heuristics, and so on, are still relatively challenging to incorporate in CIs disaster resilience models. The primary reason behind this limitation is that the majority of existing CIs resilience modeling approaches can achieve only relatively low levels of detail and granularity when modeling the operations and management processes of interdependent CIs [95], presenting a major technical gap that needs to be addressed. Prior studies have demonstrated that CIs models with higher levels of detail and granularity are better at capturing systemic heterogeneities and simulating detailed behaviors of interdependent systems during disaster scenarios such as cascading failures [96,182]. Nevertheless, the computational constraints and modeling costs in developing such models are crucial factors that currently inhibit their adoption in CIs disaster resilience modeling.

In light of the above gaps, future studies should pay particular attention to analyzing the logical interdependencies of CIs, and developing more detailed and granular disaster resilience models that can accurately capture and simulate multi-dimensional interdependencies among CIs. By so doing, future studies will be able to unveil new knowledge on unforeseen influence mechanisms across interdependent socio-technical CIs such as the ripple effects of disasters in complex system-of-systems. A large amount of data about the complex, heterogeneous and interconnected behaviors of humans and systems will also be created, which could help break the reliance on empirical data from real-life case studies for CIs disaster resilience research.

5.3. Incorporating human factors in CIs modeling: From normative models to behavioral models

Following the literature review, it was observed that until recently, the majority of studies on the human factor in the disaster resilience of CIs adopt models based on normative theories such as the expected utility theory. These studies do not address one of the main characteristics of human beings, which is their limited rationality [141]. Normative theories assume that human decision-making or behavior follows a set of norms with relatively easily predictable outcomes as long as the preconditions of the decisions can be identified. This is usually not the case in reality, where CIs decision-makers driven by personal goals, preferences, biases, aversions, and other behavioral regularities, may exhibit behaviors that deviate from normative predictions [136]. These behaviors are hard to predict since they are subjective, and their effects on the operations and management of CIs have largely remained unexplored. Moreover, normative models are designed to find the most optimal solution of decision-making problems even if these solutions may not be the most attractive to decision-makers in real-life scenarios. Disaster events such as the recent COVID-19 pandemic demonstrate that system users can also exhibit strong emergent behaviors in the face of

disasters [19]. However, it is still largely unclear what the triggering factors and effects of such emergent behaviors are. In sum, to improve the disaster resilience management of socio-technical CIs, certain knowledge gaps on the complexities of human behaviors in disaster scenarios still need to be filled.

The study of human behaviors in CIs disaster resilience is an important topic that needs to be addressed from a multidisciplinary perspective combining knowledge, techniques and tools from both engineering and social science disciplines. Lessons can be learned from other research fields on how to improve the modeling of human behavior. For example, in the broader field of behavioral operations research, the development of more accurate mathematical representations of human behavior has been a major concern among academics. This has led to the emergence of more advanced behavioral models for decision-making [136,141] that consider a wide range of behavioral regularities and social dynamics. Leveraging the knowledge and techniques from behavioral operations research and other related fields could contribute to advancing the state of the art in the subject area of this review. However, integrating knowledge and techniques from multiple disciplines presents several technical challenges. First, the heterogeneous nature of the data and models from different disciplines significantly affect their interoperability, making it difficult to develop complex simulation models. Nevertheless, with the rapid advancement in hybrid modeling and simulation techniques such as high-level architecture (HLA) based co-simulation, the possibility of integrating and achieving the interoperability of knowledge, techniques and tools from diverse disciplines is gaining some attention [95]. Second, new modeling techniques based on multidisciplinary knowledge may be challenging to validate. The multiple case specific assumptions that need to be made to facilitate the knowledge integration may also result in the limited generalizability and applicability of these techniques. Researchers have attempted to overcome these challenges by testing their models under multiple cases or using different sets of hypothetical data. Although it not always be reliable, this method can help identify common patterns in simulation results that are useful for making reasonable deductions.

In light of the above gaps, future studies should aim to deepen the knowledge on the complex behaviors of CIs stakeholders in disaster scenarios by leveraging more advanced behavioral theories and models available in other engineering, management, and social science disciplines. The novel CIs disaster resilience models that will result from integrating multidisciplinary knowledge and techniques will be able to more accurately simulate the emergent behaviors of systems and humans. As a result, a strong foundation will be laid for designing more efficient resilience management strategies and policies for safer and more sustainable cities. Moreover, these novel CIs disaster resilience models will also help unravel the complex mechanisms underlying the adaptation capacity of urban systems and societies faced with disasters. Last but not least, advanced human factor-incorporated CIs resilience modeling can help identify the aspects of CIs operations and management that can be further digitalized and automated to minimize the undesired effects of the irrational and error-prone nature of humans.

5.4. Understanding the bi-directional influence mechanism between the human and technical aspects of CIs

Socio-technical systems involve the continuous interaction and mutual influence between people and technology [20]. Although the review of literature shows that a growing amount of efforts is being made to analyze the human-system influences, the complex feedback loops between humans and systems are not sufficiently captured and explained in prior studies, presenting another major knowledge gap in the reviewed subject area that is yet to be fully addressed. Humans and systems continuously evolve throughout the lifecycle of disaster events. Humans adapt their behaviors based on past experiences and current conditions, while systems behaviors change in response to the disaster

impacts on system components and the response and recovery activities of humans. For example, decision-makers are likely to adjust their preferences and decision heuristics based on the performance evaluation of their earlier decisions [138]. Understanding the hidden feedback loops between humans and systems is of significant importance to the resilience management of CIs, as it could help identify the critical influence paths that need to be controlled to improve CIs resilience.

The above knowledge is crucial for developing disaster resilience models that can more accurately simulate the domino effects of disasters across the social and technical dimensions of CIs, and assess the direct and indirect impacts of disasters on systems and societies. The technical requirements in terms of details, granularity and data integrity for such models would be considerably higher than those discussed in the previous subsections [95]. In particular, the temporal granularity of these models would have to be significantly high to avoid any data losses or delayed responses between systems and humans that would disrupt the simulation of feedback loops between them. With the rapid advancements in computer science and the increasing capabilities of computational engines, researchers are moving closer to breaking the above technical barriers and achieving a breakthrough in the analysis and control of complex bi-directional influence mechanisms between humans and systems.

6. Conclusions

CIs are socio-technical systems that are managed and operated by humans, and hence, the limitations in human cognitive abilities, reliability, and rationality can potentially be devastating to the safety, reliability, and resilience of CIs. In this paper, a systematic review of the state of the art on the human factor in the disaster resilience modeling of CIs was conducted, approached from the angle of three phases of CIs resilience namely pre-disaster planning, emergency response, and recovery. The review of literature provided answers to the key questions that drove the need for this study. Specifically, the review helped to identify the main human factors affecting CIs resilience and classify them under individual, situational, organizational, and social factors. Moreover, several gaps and potential future research directions were identified and discussed, which included: the need for more thorough research on the resilience management of socio-technical CIs considering the multi-dimensional interdependencies between them; the need for a clearer understanding of the complexities and implications of considering the limited rationality of humans; the need to adopt a multidisciplinary perspective for the resilience modeling of CIs, one that integrates knowledge, theories and techniques from both engineering and social science disciplines; the need to model and reveal the hidden feedback loops existing between the human factors and CIs resilience; and other possible research directions about CIs resilience that can be explored using advanced human factor-incorporated CIs resilience modeling in the future. The informed discussions are expected to inspire more studies that may lead to further breakthroughs in this area.

CRedit authorship contribution statement

Joseph Jonathan Magoua: Writing – original draft, Methodology, Investigation, Conceptualization. **Nan Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Rinaldi SM, Peerenboom JP, Kelly TK. Identifying, understanding, and analyzing critical infrastructure interdependencies. *IEEE Control Syst Mag* 2001;21(6): 11–25. <https://doi.org/10.1109/37.969131>.
- [2] Kammouh O, Nogal M, Binnekamp R, Wolfert ARM. Multi-system intervention optimization for interdependent infrastructure. *Autom Constr* 2021;127:103698. <https://doi.org/10.1016/j.autcon.2021.103698>.
- [3] Timashev SA. Infrastructure resilience: definition, calculation, application. In: Proceedings of the 2015 international conference on interactive collaborative learning (ICLI); 2015. p. 1075–8. <https://doi.org/10.1109/ICLI.2015.7318180>.
- [4] Ottens M, Franssen M, Kroes P, Van De Poel I. Modelling infrastructures as socio-technical systems. *Int J Crit Infrastruct* 2006;2(2-3):133–45. <https://doi.org/10.1504/IJCIS.2006.009433>.
- [5] Wang Z, et al. Impacts of operators' behavior on reliability of power grids during cascading failures. *IEEE Trans Power Syst* 2018;33(6):6013–24. <https://doi.org/10.1109/TPWRS.2018.2825348>.
- [6] Sheridan TB. Risk, human error, and system resilience: fundamental ideas. *Hum Factors* 2008;50(3):418–26. <https://doi.org/10.1016/j.humf.2008.05.007>.
- [7] US-Canada Power System Outage Task Force. Blackout 2003: final report on the August 14, 2003 blackout in the United States and Canada: causes and recommendations. Washington, DC: Office of Electricity Delivery & Energy Reliability; 2004.
- [8] Union for the Co-ordination of Electricity Transmission (UCTE). Interim report of the investigation committee on the 28 September 2003 blackout in Italy. Brussels, Belgium: Union for the Co-ordination of Electricity Transmission (UCTE); 2003.
- [9] Sun Z, Zhang C, Tang P. Simulation-based optimization of communication protocols for reducing delays during nuclear power plant outages. In: Proceedings of the construction research congress 2018; 2018. p. 455–64. <https://doi.org/10.1061/9780784481295.046>.
- [10] Bao Y, Guo C, Zhang J, Wu J, Pang S, Zhang Z. Impact analysis of human factors on power system operation reliability. *J Modern Power Syst Clean Energy* 2018;6(1):27–39. <https://doi.org/10.1007/s40565-016-0231-6>.
- [11] Nan C, Sansavini G. Developing an agent-based hierarchical modeling approach to assess human performance of infrastructure systems. *Int J Ind Ergon* 2016;53: 340–54. <https://doi.org/10.1016/j.ergon.2016.04.002>.
- [12] Zarei E, Khan F, Abbassi R. Importance of human reliability in process operation: a critical analysis. *Reliab Eng Syst Saf* 2021;211:107607. <https://doi.org/10.1016/j.res.2021.107607>.
- [13] Bolton ML, Zheng X, Kang E. A formal method for including the probability of erroneous human task behavior in system analyses. *Reliab Eng Syst Saf* 2021;213: 107764. <https://doi.org/10.1016/j.res.2021.107764>.
- [14] Balci B, Silvestri S, Rancourt ME, Laporte G. Collaborative prepositioning network design for regional disaster response. *Prod Oper Manag* 2019;28(10): 2431–55. <https://doi.org/10.1111/poms.13053>.
- [15] Naderpajouh N, Yu DJ, Aldrich DP, Linkov I, Matinheikki J. Engineering meets institutions: an interdisciplinary approach to the management of resilience. *Environ Syst Decis* 2018;38(3):306–17. <https://doi.org/10.1007/s10669-018-9704-7>.
- [16] Ramaswami A, et al. A social-ecological-infrastructure systems framework for interdisciplinary study of sustainable city systems. *J Ind Ecol* 2012;16(6):801–13. <https://doi.org/10.1111/j.1530-9290.2012.00566.x>.
- [17] Barry A. The material politics of infrastructure. In: Maassen S, Dickel S, Schneider C, editors. *TechnoScienceSociety: technological reconfigurations of science and society*. Cham: Springer International Publishing; 2020. p. 91–109.
- [18] Frank B, Delano D, Caniglia B. Urban systems: a socio-ecological system perspective. *Sociol Int J* 2017;1. <https://doi.org/10.15406/sij.2017.01.00001>.
- [19] Kim K. Impacts of COVID-19 on transportation: summary and synthesis of interdisciplinary research. In: *eng Transp Res Interdiscip Perspect* 2021;9:100305. <https://doi.org/10.1016/j.trip.2021.100305>.
- [20] Chester M, et al. Infrastructure resilience to navigate increasingly uncertain and complex conditions in the Anthropocene. *NPJ Urban Sustain* 2021;1(1):4. <https://doi.org/10.1038/s42949-021-00016-y>.
- [21] Ouyang M. Review on modeling and simulation of interdependent critical infrastructure systems. *Reliab Eng Syst Saf* 2014;121:43–60. <https://doi.org/10.1016/j.res.2013.06.040>.
- [22] Ackerman D, Mehrpouyan H. Modeling human behavior to anticipate insider attacks via system dynamics. In: Proceedings of the 2016 symposium on theory of modeling and simulation (TMS-DEVS); 2016. p. 1–6. <https://doi.org/10.23919/TMS.2016.7918809>.
- [23] Kyriakidis M, Majumdar A, Ochieng WY. The human performance railway operational index—a novel approach to assess human performance for railway operations. *Reliab Eng Syst Saf* 2018;170:226–43. <https://doi.org/10.1016/j.res.2017.10.012>.
- [24] Li Y, Mosleh A. Modeling and simulation of crew to crew response variability due to problem-solving styles. *Reliab Eng Syst Saf* 2020;194:105840. <https://doi.org/10.1016/j.res.2017.05.020>.
- [25] Hunt K, Agarwal P, Zhuang J. Technology adoption for airport security: modeling public disclosure and secrecy in an attacker-defender game. *Reliab Eng Syst Saf* 2021;207:107355. <https://doi.org/10.1016/j.res.2020.107355>.
- [26] Hassan EM, Mahmoud H. An integrated socio-technical approach for post-earthquake recovery of interdependent healthcare system. *Reliab Eng Syst Saf* 2020;201:106953. <https://doi.org/10.1016/j.res.2020.106953>.
- [27] Reilly AC, Baroud H, Flage R, Gerst MD. Sources of uncertainty in interdependent infrastructure and their implications. *Reliab Eng Syst Saf* 2021;213:107756. <https://doi.org/10.1016/j.res.2021.107756>.
- [28] Bauer JM, Herder PM. Designing Socio-Technical Systems. In: Meijers A, editor. *Philosophy of technology and engineering sciences*. Amsterdam: North-Holland; 2009. p. 601–30.
- [29] Osman H, Ali MRH. Complex systems modeling of infrastructure assets, operators, users, and politicians using system dynamics. In: Proceedings of the construction research congress. 2012; 2012. p. 2280–9.
- [30] Trump BD, et al. Social resilience and critical infrastructure systems. In: Linkov I, Palma-Oliveira JM, editors. *Resilience and risk*. Dordrecht: Springer Netherlands; 2017. p. 289–99.
- [31] Muir A, Lopatto J. Final report on the August 14 2003 blackout in the United States and Canada: causes and recommendations. US - Canada Power System Outage Task Force 2004:1–238.
- [32] Holmberg JE, Kahlbom U. Application of human reliability analysis in the deterministic safety analysis for nuclear power plants. *Reliab Eng Syst Saf* 2020; 194:106371. <https://doi.org/10.1016/j.res.2019.02.003>.
- [33] Sun Z, Xing J, Tang P, Cooke NJ, Boring RL. Human reliability for safe and efficient civil infrastructure operation and maintenance – a review. *Dev Built Environ* 2020;4:100028. <https://doi.org/10.1016/j.dibe.2020.100028>.
- [34] Endsley MR. Toward a theory of situation awareness in dynamic systems. *Hum Factors* 1995;37(1):32–64. <https://doi.org/10.1016/001872095779049543>.
- [35] Morais C, Yung KL, Johnson K, Moura R, Beer M, Patelli E. Identification of human errors and influencing factors: a machine learning approach. *Saf Sci* 2022; 146:105528. <https://doi.org/10.1016/j.ssci.2021.105528>.
- [36] Diller T, Helmrich G, Dunning S, Cox S, Buchanan A, Shappell S. The human factors analysis classification system (HFACS) applied to health care. *Am J Med Qual* 2013;29(3):181–90. <https://doi.org/10.1177/1062860613491623>.
- [37] Esmalian A, Dong S, Mostafaei A. Susceptibility curves for humans: empirical survival models for determining household-level disturbances from hazards-induced infrastructure service disruptions. *Sustain Cities Soc* 2021;66:102694. <https://doi.org/10.1016/j.scs.2020.102694>.
- [38] Franchina L, et al. Passive and active training approaches for critical infrastructure protection. *Int J Disaster Risk Reduct* 2021;63:102461. <https://doi.org/10.1016/j.ijdrr.2021.102461>.
- [39] Pagano A, Pluchinotta I, Giordano R, Vurro M. Drinking water supply in resilient cities: notes from L'Aquila earthquake case study. *Sustain Cities Soc* 2017;28: 435–49. <https://doi.org/10.1016/j.scs.2016.09.005>.
- [40] Leong C. The role of narratives in sociohydrological models of flood behaviors. *Water Resour Res* 2018;54(4):3100–21. <https://doi.org/10.1002/2017WR022036>.
- [41] Alves PBR, Djordjević S, Javadi AA. Addressing social and institutional vulnerabilities in the context of flood risk mitigation. *J Flood Risk Manag* 2022; e12839. <https://doi.org/10.1111/jfr.12839>.
- [42] Nelson PP. A framework for the future of urban underground engineering. *Tunnel Undergr Space Technol* 2016;55:32–9. <https://doi.org/10.1016/j.tust.2015.10.023>.
- [43] Ongkwojjoyo C, Doloi H. Determining critical infrastructure risks using social network analysis. *Int J Disaster Resil Built Environ* 2017;8(1):5–26. <https://doi.org/10.1108/IJDRBE-05-2016-0016>.
- [44] Zhang C, et al. Human-centered automation for resilient nuclear power plant outage control. *Autom Constr* 2017;82:179–92. <https://doi.org/10.1016/j.autcon.2017.05.001>.
- [45] Duan S, Ayyub BM. Assessment methods of network resilience for cyber-human-physical systems. *ASCE-ASME J Risk Uncertainty Eng Syst Part A Civil Eng* 2020; 6(1):03119001. <https://doi.org/10.1061/AJRU6.0001021>.
- [46] Brauner F, Claßen M, Fiedrich F. Competence as enabler of urban critical infrastructure resilience assessment. In: Fekete A, Fiedrich F, editors. *Urban disaster resilience and security: addressing risks in societies*. Cham: Springer International Publishing; 2018. p. 171–84.
- [47] Lo JC, Pluyter KR, Meijer SA. Individual markers of resilience in train traffic control: the role of operators' goals and strategic mental models and implications for variation, expertise, and performance. *Hum Factors* 2016;58(1):80–91. <https://doi.org/10.1177/0018720815622357>.
- [48] Mitsova D. Integrative interdisciplinary approaches to critical infrastructure interdependency analysis. *Risk Anal* 2021;41(7):1111–7. <https://doi.org/10.1111/risa.13129>.
- [49] Achilloupolou DV, Mitoulis SA, Argyroudis SA, Wang Y. Monitoring of transport infrastructure exposed to multiple hazards: a roadmap for building resilience. *Sci Total Environ* 2020;746:141001. <https://doi.org/10.1016/j.scitotenv.2020.141001>.
- [50] Heglund J, Hopkinson KM, Tran HT. Social sensing: towards social media as a sensor for resilience in power systems and other critical infrastructures. *Sustain*

- Resilient Infrastruct 2021;6(1-2):94–106. <https://doi.org/10.1080/23789689.2020.1719728>.
- [51] Dev A, Dilly TC, Bakhshpour AE, Dittmer U, Bhallamudi SM. Optimal implementation of wastewater reuse in existing sewerage systems to improve resilience and sustainability in water supply systems. *Water* 2021;13(15):2004.
 - [52] Shuvro RA, Das P, Abreu JM, Hayat MM. Correlating grid-operators' performance with cascading failures in smart-grids. In: Proceedings of the 2019 IEEE PES innovative smart grid technologies Europe (ISGT-Europe); 2019. p. 1–5. <https://doi.org/10.1109/ISGTEurope.2019.8905687>.
 - [53] Buddaraju D. Performance of control room operators in alarm management. Louisiana State University; 2011. p. 1105. Master's Thesis.
 - [54] Bouarfa S, Blom HAP, Curran R, Everdij MHC. Agent-based modeling and simulation of emergent behavior in air transportation. *Complex Adapt Syst Model* 2013;1(1):15. <https://doi.org/10.1186/2194-3206-1-15>.
 - [55] Sun L, Stojadinovic B, Sansavini G. Agent-based recovery model for seismic resilience evaluation of electrified communities. in *eng Risk Anal* 2019;39(7): 1597–614. <https://doi.org/10.1111/risa.13277>.
 - [56] Bao Y, et al. Analysis of human reliability in power system switching operation considering dependency of operators. In: Proceedings of the 2013 IEEE PES Asia-Pacific power and energy engineering conference (APPEEC); 2013. p. 1–6. <https://doi.org/10.1109/APPEEC.2013.6837155>.
 - [57] Yang Y, et al. BIM-GIS-DCEs enabled vulnerability assessment of interdependent infrastructures – a case of stormwater drainage-building-road transport Nexus in urban flooding. *Autom Constr* 2021;125:103626. <https://doi.org/10.1016/j.autcon.2021.103626>.
 - [58] Petrillo A, Falcone D, De Felice F, Zomparelli F. Development of a risk analysis model to evaluate human error in industrial plants and in critical infrastructures. *Int J Disaster Risk Reduct* 2017;23:15–24. <https://doi.org/10.1016/j.ijdrr.2017.03.012>.
 - [59] Rushby J. Modeling the human in human factors. In: Voges U, editor. *Computer safety, reliability and security*. Berlin, Heidelberg: Springer Berlin Heidelberg; 2001. p. 86–91.
 - [60] Sun L, Stojadinovic B, Sansavini G. Resilience evaluation framework for integrated civil infrastructure 2013; community systems under seismic hazard. *J Infrastruct Syst* 2019;25(2):04019016. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000492](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000492).
 - [61] Lawless MT, Laughery KR, Persenky JJ. Using micro saint to predict performance in a nuclear power plant control room, report. Washington D.C: University of North Texas Libraries, UNT Digital Library, crediting UNT Libraries Government Documents Department; 1995.
 - [62] Dong S, Esmalian A, Farahmand H, Mostafavi A. An integrated physical-social analysis of disrupted access to critical facilities and community service-loss tolerance in urban flooding. *Comput Environ Urban Syst* 2020;80:101443. <https://doi.org/10.1016/j.compenvurbysys.2019.101443>.
 - [63] Matos P, Vieira J, Rocha B, Branquinho C, Pinho P. Modeling the provision of air-quality regulation ecosystem service provided by urban green spaces using lichens as ecological indicators. *Sci Total Environ* 2019;665:521–30. <https://doi.org/10.1016/j.scitotenv.2019.02.023>.
 - [64] Ge Y, Du L, Ye H. Co-optimization approach to post-storm recovery for interdependent power and transportation systems. *J Modern Power Syst Clean Energy* 2019;7(4):688–95. <https://doi.org/10.1007/s40565-019-0524-7>.
 - [65] Oikonomou K, Parvania M. Optimal coordinated operation of interdependent power and water distribution systems. *IEEE Trans Smart Grid* 2020;11(6): 4784–94. <https://doi.org/10.1109/TSG.2020.3000173>.
 - [66] Zlotnik A, Roald L, Backhaus S, Chertkov M, Andersson G. Coordinated scheduling for interdependent electric power and natural gas infrastructures. *IEEE Trans Power Syst* 2017;32(1):600–10. <https://doi.org/10.1109/TPWRS.2016.2545522>.
 - [67] Talebman H, Duenas-Osorio L. Decentralized decision making for the restoration of interdependent networks. *ASCE-ASME J Risk U A* 2020;6(2):04020012. <https://doi.org/10.1061/AJRUAA.6.0001035>.
 - [68] Mohebbi S, Barnett K, Aslani B. Decentralized resource allocation for interdependent infrastructures resilience: a cooperative game approach. *Int Trans Oper Res* 2021;28(6):3394–415. <https://doi.org/10.1111/itor.12978>.
 - [69] Akbar PhD MS, Aldrich DP. Social capital's role in recovery: evidence from communities affected by the 2010 Pakistan floods. *Disasters* 2018;42(3):475–97. <https://doi.org/10.1111/disa.12259>.
 - [70] Mohaghegh Z, Mosleh A. Incorporating organizational factors into probabilistic risk assessment of complex socio-technical systems: principles and theoretical foundations. *Saf Sci* 2009;47(8):1139–58. <https://doi.org/10.1016/j.ssci.2008.12.008>.
 - [71] Harrison JR, Lin Z, Carroll GR, Carley KM. Simulation modeling in organizational and management research. in *English Acad Manag Rev* 2007;32(4):1229–45.
 - [72] Aghababaei M, Koliou M. Community resilience assessment via agent-based modeling approach. *Comput Aided Civil Infrastruct Eng* 2022;38:1–20. <https://doi.org/10.1111/mice.12916>.
 - [73] Zimmerman R. Decision-making and the vulnerability of interdependent critical infrastructure. In: Proceedings of the 2004 IEEE international conference on systems, man and cybernetics (IEEE Cat. No.04CH37583). 5; 2004. p. 4059–63. <https://doi.org/10.1109/ICSMC.2004.1401166>. vol.5.
 - [74] Rome E, Langeslag P, Usov A. Federated modelling and simulation for critical infrastructure protection. In: D'Agostino G, Scala A, editors. *Networks of networks: the last frontier of complexity*. Cham: Springer International Publishing; 2014. p. 225–53.
 - [75] Anokhin D, Dehghanian P, Lejeune MA, Su J. Mobility-as-a-service for resilience delivery in power distribution systems. *Prod Oper Manag* 2021;30(8):2492–521. <https://doi.org/10.1111/poms.13393>.
 - [76] Wang T, Liu B, Zhang J, Li G. A real options-based decision-making model for infrastructure investment to prevent rainstorm disasters. *Prod Oper Manag* 2019; 28(11):2699–715. <https://doi.org/10.1111/poms.13074>.
 - [77] Dao J, Ng ST, Yang Y, Zhou S, Xu FJ, Skitmore M. Semantic framework for interdependent infrastructure resilience decision support. *Autom Constr* 2021; 130:103852. <https://doi.org/10.1016/j.autcon.2021.103852>.
 - [78] Hunt JCR, Timoshkina Y, Baudains PJ, Bishop SR. System dynamics applied to operations and policy decisions. *Eur Rev* 2012;20(3):324–42. <https://doi.org/10.1017/S1062798711000585>.
 - [79] Emanuel R, Ayyub B. Assessing resilience model responsiveness in the context of stakeholder preferences in decision support systems. *ASCE-ASME J Risk Uncertainty Eng Syst Part A Civil Eng* 2019;5(2):04019005. <https://doi.org/10.1061/AJRUAA.6.0001002>.
 - [80] Lianxiao TM, Siqin T, An C, Yuhai B, Fei G. Evaluating regional flood disaster resilience based on the DROP model and TOPSIS method: a case study of the Katsushika Ward, Tokyo. *Nat Hazards Rev* 2022;23(3):04022018. [https://doi.org/10.1061/\(ASCE\)NH.1527-6996.0000551](https://doi.org/10.1061/(ASCE)NH.1527-6996.0000551).
 - [81] Almutairi A, Thorisson H, Wheeler JP, Slutsky DL, Lambert JH. Scenario-based preferences in development of advanced mobile grid services and a bidirectional charger network. *ASCE-ASME J Risk Uncertainty Eng Syst Part A Civil Eng* 2018;4 (2):04018017. <https://doi.org/10.1061/AJRUAA.6.0000962>.
 - [82] Gomez C, González AD, Baroud H, Bedoya-Motta CD. Integrating operational and organizational aspects in interdependent infrastructure network recovery. *Risk Anal* 2019;39(9):1913–29. <https://doi.org/10.1111/risa.13340>.
 - [83] Almutairi A, Wheeler JP, Slutsky DL, Lambert JH. Integrating stakeholder mapping and risk scenarios to improve resilience of cyber-physical-social networks. *Risk Anal* 2019;39(9):2093–112. <https://doi.org/10.1111/risa.13292>.
 - [84] Gumiran BA, Daag A. Negotiated participatory action research for multi-stakeholder implementation of early warning systems for landslides. *Int J Disaster Risk Reduct* 2021;58:102184. <https://doi.org/10.1016/j.ijdrr.2021.102184>.
 - [85] Saikia P, et al. City water resilience framework: a governance based planning tool to enhance urban water resilience. *Sustain Cities Soc* 2022;77:103497. <https://doi.org/10.1016/j.scs.2021.103497>.
 - [86] Hart SD, Klosky JL, Katalenich S. Conceptual models for infrastructure leadership. *J Manag Eng* 2014;30(3):04014003. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000217](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000217).
 - [87] Ashmore FH, Farrington JH, Skerratt S. Community-led broadband in rural digital infrastructure development: implications for resilience. *J Rural Stud* 2017;54: 408–25. <https://doi.org/10.1016/j.jrurstud.2016.09.004>.
 - [88] Koh ECY. Resilience analysis of infrastructure systems in incremental design change. *Comput Ind* 2022;142:103734. <https://doi.org/10.1016/j.compind.2022.103734>.
 - [89] Didier M, Broccardo M, Esposito S, Stojadinovic B. A compositional demand/supply framework to quantify the resilience of civil infrastructure systems (Re-CoDeS). *Sustain Resilient Infrastruct* 2018;3(2):86–102. <https://doi.org/10.1080/23789689.2017.1364560>.
 - [90] Steffen W, Broadgate W, Deutsch L, Gaffney O, Ludwig C. The trajectory of the anthropocene: the great acceleration. *The Anth Rev* 2015;2(1):81–98. <https://doi.org/10.1177/2053019614564785>.
 - [91] Öst J, Dolciotti M, Reggiani A, Nijkamp P. Social capital, resilience and accessibility in urban systems: a study on Sweden. *Netw Spatial Econ* 2018;18(2): 313–36. <https://doi.org/10.1007/s11067-017-9375-9>.
 - [92] Aldrich DP, Meyer MA. Social capital and community resilience. *Am Behav Sci* 2015;59(2):254–69. <https://doi.org/10.1177/0002764214550299>.
 - [93] Dhakal S, Zhang L, Lv X. Understanding infrastructure resilience, social equity, and their interrelationships: exploratory study using social media data in Hurricane Michael. *Nat Hazards Rev* 2021;22(4):04021045. [https://doi.org/10.1061/\(ASCE\)NH.1527-6996.0000512](https://doi.org/10.1061/(ASCE)NH.1527-6996.0000512).
 - [94] Meerow S, Pajouhesh P, Miller TR. Social equity in urban resilience planning. *Local Environ* 2019;24(9):793–808. <https://doi.org/10.1080/13549839.2019.1645103>.
 - [95] Magoua JJ, Wang F, Li N. High level architecture-based framework for modeling interdependent critical infrastructure systems. *Simul Model Pract Theory* 2022: 102529. <https://doi.org/10.1016/j.simpat.2022.102529>.
 - [96] Wang F, Magoua JJ, Li N. Modeling cascading failure of interdependent critical infrastructure systems using HLA-based co-simulation. *Autom Constr* 2022;133: 104008. <https://doi.org/10.1016/j.autcon.2021.104008>.
 - [97] Trucco P, Cagno E, De Ambroggi M. Dynamic functional modelling of vulnerability and interoperability of critical infrastructures. *Reliab Eng Syst Saf* 2012;105:51–63. <https://doi.org/10.1016/j.res.2011.12.003>.
 - [98] Faust KM, Abraham DM, DeLaurentis D. Coupled human and water infrastructure systems sector interdependencies: framework evaluating the impact of cities experiencing urban decline. *J Water Resour Plan Manag* 2017;143(8):04017043. [https://doi.org/10.1061/\(ASCE\)WJWR.1943-5452.0000794](https://doi.org/10.1061/(ASCE)WJWR.1943-5452.0000794).
 - [99] Sadri AM, et al. The role of social capital, personal networks, and emergency responders in post-disaster recovery and resilience: a study of rural communities in Indiana. *Nat Hazards* 2018;90(3):1377–406. <https://doi.org/10.1007/s11069-017-3103-0>.
 - [100] Ulak MB, Yazici A, Ozguven EE. A prescriptive model to assess the socio-demographics impacts of resilience improvements on power networks. *Int J Disaster Risk Reduct* 2020;51:101777. <https://doi.org/10.1016/j.ijdrr.2020.101777>.

- [101] Esmalian A, Wang W, Mostafavi A. Multi-agent modeling of hazard-household-infrastructure nexus for equitable resilience assessment. *Comput Aided Civil Infrastruct Eng* 2022;37(12):1491–520. <https://doi.org/10.1111/mice.12818>.
- [102] Patterson O, Weil F, Patel K. The role of community in disaster response: conceptual models. *Popul Res Policy Rev* 2010;29(2):127–41. <https://doi.org/10.1007/s11113-009-9133-x>.
- [103] Twumasi-Boakye R, Sobanjo JO. Resilience of regional transportation networks subjected to hazard-induced bridge damages. *J Trans Eng Part A Syst* 2018;144(10):04018062. <https://doi.org/10.1061/JTEPBS.0000186>.
- [104] He X, Cha EJ. Modeling the damage and recovery of interdependent critical infrastructure systems from natural hazards. *Reliab Eng Syst Saf* 2018;177:162–75. <https://doi.org/10.1016/j.res.2018.04.029>.
- [105] Kong J, Zhang C, Simonovic SP. A two-stage restoration resource allocation model for enhancing the resilience of interdependent infrastructure systems. *Sustainability* 2019;11(19):5143.
- [106] Smith AM, González AD, Dueñas-Osorio L, D'Souza RM. Interdependent network recovery games. *Risk Anal* 2020;40(1):134–52. <https://doi.org/10.1111/risa.12923>.
- [107] Pandit A, Crittenden JC. Index of network resilience for urban water distribution systems. *Int J Crit Infrastruct* 2016;12(1–2):120–42. <https://doi.org/10.1504/IJCIS.2016.075865>.
- [108] Kushal TRB, Illindala MS. Decision support framework for resilience-oriented cost-effective distributed generation expansion in power systems. *IEEE Trans Ind Appl* 2021;57(2):1246–54. <https://doi.org/10.1109/TIA.2020.3047595>.
- [109] Zhang W, Wang N, Nicholson C. Resilience-based post-disaster recovery strategies for road-bridge networks. *Struct Infrastruct Eng* 2017;13(11):1404–13. <https://doi.org/10.1080/15732479.2016.1271813>.
- [110] Sarkale Y, Nozhati S, Chong EKP, Ellingwood BR, Mahmoud H. Solving Markov decision processes for network-level post-hazard recovery via simulation optimization and rollout. In: *Proceedings of the 2018 IEEE 14th international conference on automation science and engineering (CASE)*; 2018. p. 906–12. <https://doi.org/10.1109/COASE.2018.8560473>.
- [111] Niu S, Zhang JS, Zhang F, Gao J. Resilience analysis for comprehensive transportation network. In: Wang W, Baumann M, Jiang X, editors. *Green, smart and connected transportation systems*. Singapore: Springer Singapore; 2020. p. 195–205.
- [112] Shuvro RA, Wang Z, Das P, Naeini MR, Hayat MM. Modeling impact of communication network failures on power grid reliability. In: *Proceedings of the 2017 North American power symposium (NAPS)*; 2017. p. 1–6. <https://doi.org/10.1109/NAPS.2017.8107224>.
- [113] Nakarmi U, Rahnamay-Naeini M. A Markov chain approach for cascade size analysis in power grids based on community structures in interaction graphs. In: *Proceedings of the 2020 international conference on probabilistic methods applied to power systems (PMAPS)*; 2020. p. 1–6. <https://doi.org/10.1109/PMAPS47429.2020.9183579>.
- [114] Memarzadeh M, Pozzi M. Integrated inspection scheduling and maintenance planning for infrastructure systems. *Comput Aided Civil Infrastruct Eng* 2016;31(6):403–15. <https://doi.org/10.1111/MICE.12178>.
- [115] Liu Y, Chen Y, Jiang T. Dynamic selective maintenance optimization for multi-state systems over a finite horizon: a deep reinforcement learning approach. *Eur J Oper Res* 2020;283(1):166–81. <https://doi.org/10.1016/j.ejor.2019.10.049>.
- [116] Rashedi N, Tajeddini MA, Kebraei H. Markov game approach for multi-agent competitive bidding strategies in electricity market. *IET Gen Trans Distrib* 2016;10(15):3756–63. <https://doi.org/10.1049/iet-gtd.2016.0075>.
- [117] Anderson Jr EG, Lewis K, Ozer GT. Combining stock-and-flow, agent-based, and social network methods to model team performance. *Syst Dyn Rev* 2018;34(4):527–74. <https://doi.org/10.1002/sdr.1613>.
- [118] Hsu JC, Clymer JR, Garcia Jr J, Gonzalez E. Agent-based modeling the emergent behavior of a system-of-systems. In: *Proceedings of the INCOSE international symposium*. 19; 2009. p. 1581–90. <https://doi.org/10.1002/j.2334-5837.2009.tb01036.x>.
- [119] Kolen S, Dähling S, Isermann T, Monti A. Enabling the analysis of emergent behavior in future electrical distribution systems using agent-based modeling and simulation. *Complexity* 2018;2018:1–16. <https://doi.org/10.1155/2018/3469325>.
- [120] Johansen C, Tien I. Probabilistic multi-scale modeling of interdependencies between critical infrastructure systems for resilience. *Sustain Resilient Infrastruct* 2018;3(1):1–15. <https://doi.org/10.1080/23789689.2017.1345253>.
- [121] Hossain NUI, Jaradat R, Hosseini S, Marufuzzaman M, Buchanan RK. A framework for modeling and assessing system resilience using a Bayesian network: A case study of an interdependent electrical infrastructure system. *Int J Crit Infrastruct Prot* 2019;25:62–83. <https://doi.org/10.1016/j.ijcip.2019.02.002>.
- [122] Hossain NUI, Nur F, Hosseini S, Jaradat R, Marufuzzaman M, Puryear SM. A Bayesian network based approach for modeling and assessing resilience: A case study of a full service deep water port. *Reliab Eng Syst Saf* 2019;189:378–96. <https://doi.org/10.1016/j.res.2019.04.037>.
- [123] Hu Y, Li R, Zhang Y. Predicting pilot behavior during midair encounters using recognition primed decision model. *Inf Sci* 2018;422:377–95. <https://doi.org/10.1016/j.ins.2017.09.035>.
- [124] Bowman W, Messner M, Regli S, Bender J. Measuring the effectiveness of performance-based training. *J Water Health* 2008;7(1):155–67. <https://doi.org/10.2166/wh.2009.004>.
- [125] Liu X, Ferrario E, Zio E. Identifying resilient-important elements in interdependent critical infrastructures by sensitivity analysis. *Reliab Eng Syst Saf* 2019;189:423–34. <https://doi.org/10.1016/j.res.2019.04.017>.
- [126] Dai D, Sun M, Lv X, Lei K. Evaluating water resource sustainability from the perspective of water resource carrying capacity, a case study of the Yongding River watershed in Beijing-Tianjin-Hebei region, China. *Environ Sci Pollut Res* 2020;27(17):21590–603. <https://doi.org/10.1007/s11356-020-08259-5>.
- [127] Lei S, et al. Cognitive abilities predict safety performance: a study examining high-speed railway dispatchers. *J Adv Trans* 2021;2021:5538320. <https://doi.org/10.1155/2021/5538320>.
- [128] Cavallini S, et al. A system dynamics framework for modeling critical infrastructure resilience. *IFIP Adv Inf Commun Technol* 2014;441:141–54.
- [129] Goldbeck N, Angeloudis P, Ochieng WY. Resilience assessment for interdependent urban infrastructure systems using dynamic network flow models. *Reliab Eng Syst Saf* 2019;188:62–79. <https://doi.org/10.1016/j.res.2019.03.007>.
- [130] Abdul Rahman F, Varuttamaseni A, Kintner-Meyer M, Lee JC. Application of fault tree analysis for customer reliability assessment of a distribution power system. *Reliab Eng Syst Saf* 2013;111:76–85. <https://doi.org/10.1016/j.res.2012.10.011>.
- [131] Hussain S, Lee KB, Ahmed MA, Hayes B, Kim YC. Two-stage fuzzy logic inference algorithm for maximizing the quality of performance under the operational constraints of power grid in electric vehicle parking lots. *Energies* 2020;13(18):4634.
- [132] MacKenzie CA, Baroud H, Barker K. Static and dynamic resource allocation models for recovery of interdependent systems: application to the deepwater horizon oil spill. *Ann Oper Res* 2016;236(1):103–29. <https://doi.org/10.1007/s10479-014-1696-1>.
- [133] Zhang C, Kong JJ, Simonovic SP. Restoration resource allocation model for enhancing resilience of interdependent infrastructure systems. *Saf Sci* 2018;102:169–77. <https://doi.org/10.1016/j.ssci.2017.10.014>.
- [134] Çağnan Z, Davidson RA, Guikema SD. Post-earthquake restoration planning for Los Angeles electric power. *Earthq Spectra* 2006;22(3):589–608. <https://doi.org/10.1193/1.2222400>.
- [135] Xu N, Guikema SD, Davidson RA, Nozick LK, Çağnan Z, Vaziri K. Optimizing scheduling of post-earthquake electric power restoration tasks. *Earthq Eng Struct Dyn* 2007;36(2):265–84. <https://doi.org/10.1002/eqe.623>.
- [136] Morrison JB, Oliva R. Integration of behavioral and operational elements through system dynamics. *The Handbook of Behavioral Operations* 2018:287–321.
- [137] Krishnamurti T, et al. Preparing for smart grid technologies: a behavioral decision research approach to understanding consumer expectations about smart meters. *Energy Policy* 2012;41:790–7. <https://doi.org/10.1016/j.enpol.2011.11.047>.
- [138] Federal Emergency Management Agency. *National pre-disaster mitigation fund. Report to Congress Fiscal Year 2017*:1–14.
- [139] Hu J, Dong S, Zhang L, Chen Y, Xu K. Cyber-physical-social hazard analysis for LNG port terminal system based on interdependent network theory. *Saf Sci* 2021;137:105180. <https://doi.org/10.1016/j.ssci.2021.105180>.
- [140] Gravette MA, Barker K. Achieved availability importance measure for enhancing reliability-centered maintenance decisions. *Proc Inst Mech Eng Part J Risk Reliab* 2014;229(1):62–72. <https://doi.org/10.1177/1748006x14550849>.
- [141] Carpinelli G, Russo A. Behavioral perspective of power systems' decision makers. *Int J Electr Power Energy Syst* 2014;58:111–9. <https://doi.org/10.1016/j.ijepes.2014.01.002>.
- [142] Scholz RW. Introduction to decision making under uncertainty: biases, fallacies, and the development of decision making. *Adv Psychol* 1983;16(1):3–18.
- [143] Tversky A, Kahneman D. Judgment under uncertainty: heuristics and biases. *Science* 1974;185(4157):1124–31. <https://doi.org/10.1126/science.185.4157.1124>.
- [144] Marx SM, et al. Communication and mental processes: experiential and analytic processing of uncertain climate information. *Global Environ Change* 2007;17(1):47–58. <https://doi.org/10.1016/j.gloenvcha.2006.10.004>.
- [145] Masini A, Menichetti E. The impact of behavioural factors in the renewable energy investment decision making process: conceptual framework and empirical findings. *Energy Policy* 2012;40:28–38. <https://doi.org/10.1016/j.enpol.2010.06.062>.
- [146] Feng D, Xu Z. Risk analysis of volume cheat strategy in a competitive capacity market. In: *Proceedings of the 2009 IEEE power & energy society general meeting*; 2009. p. 1–7. <https://doi.org/10.1109/PES.2009.5275804>.
- [147] Bruneau M, et al. A framework to quantitatively assess and enhance the seismic resilience of communities. *Earthq Spectra* 2003;19(4):733–52. <https://doi.org/10.1193/1.1623497>.
- [148] Sharma N, Tabandeh A, Gardoni P. Resilience analysis: a mathematical formulation to model resilience of engineering systems. *Sustain Resilient Infrastruct* 2018;3(2):49–67. <https://doi.org/10.1080/23789689.2017.1345257>.
- [149] Pham Q, Hwang W. Fairness-aware spectral and energy efficiency in spectrum-sharing wireless networks. *IEEE Trans Veh Technol* 2017;66(11):10207–19. <https://doi.org/10.1109/TVT.2017.2717926>.
- [150] Haines YY, Horowitz BM, Lambert JH, Santos JR, Lian C, Crowther KG. Inoperability input-output model for interdependent infrastructure sectors. i: theory and methodology. *J Infrastruct Syst* 2005;11(2):67–79. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2005\)11:2\(67\)](https://doi.org/10.1061/(ASCE)1076-0342(2005)11:2(67)).
- [151] Fatorechi R, Miller-Hooks E. Measuring the performance of transportation infrastructure systems in disasters: a comprehensive review. *J Infrastruct Syst* 2015;21(1):04014025. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000212](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000212).
- [152] Coles J, Zhuang J. Decisions in disaster recovery operations: a game theoretic perspective on organization cooperation. *J Homel Secur Emerg Manag* 2011;8(1). <https://doi.org/10.2202/1547-7355.1772>.

- [153] Perrier N, et al. A survey of models and algorithms for emergency response logistics in electric distribution systems. Part II: contingency planning level. *Comput Oper Res* 2013;40(7):1907–22. <https://doi.org/10.1016/j.cor.2012.09.009>.
- [154] Wu J, Lee T, Cao C. Intelligent crew and outage scheduling in electrical distribution system by hybrid generic algorithm. In: *Proceedings of the 2009 4th IEEE conference on industrial electronics and applications*; 2009. p. 96–101. <https://doi.org/10.1109/ICIEA.2009.5138177>.
- [155] Guha S, Moss A, Naor J, Schieber B. Efficient recovery from power outage (extended abstract). In: *Proceedings of the thirty-first annual ACM symposium on theory of computing*; 1999.
- [156] Ryan B, Golightly D, Pickup L, Reinartz S, Atkinson S, Dadashi N. Human functions in safety - developing a framework of goals, human functions and safety relevant activities for railway socio-technical systems. *Saf Sci* 2021;140:105279. <https://doi.org/10.1016/j.ssci.2021.105279>.
- [157] Sant'Anna DALMd, Hilal AVGD. The impact of human factors on pilots' safety behavior in offshore aviation companies: a Brazilian case. *Saf Sci* 2021;140:105272. <https://doi.org/10.1016/j.ssci.2021.105272>.
- [158] Weintraub A, Aboud J, Fernandez C, Laporte G, Ramirez E. An emergency vehicle dispatching system for an electric utility in Chile. *J Oper Res Soc* 1999;50(7):690–6. <https://doi.org/10.1057/palgrave.jors.2600746>.
- [159] Ming-Jong Y, Min KJ. Repair-unit location models for power failures. *IEEE Trans Eng Manag* 1998;45(1):57–65. <https://doi.org/10.1109/17.658661>.
- [160] Fritzen PC, Zauk JM, Cardoso G, Oliveira AdL, de Araújo OCB. Hybrid system based on constructive heuristic and integer programming for the solution of problems of fault section estimation and alarm processing in power systems. *Electric Power Syst Res* 2012;90:55–66. <https://doi.org/10.1016/j.epsr.2012.04.002>.
- [161] Musharraf M, Hassan J, Khan F, Veitch B, MacKinnon S, Imtiaz S. Human reliability assessment during offshore emergency conditions. *Saf Sci* 2013;59:19–27. <https://doi.org/10.1016/j.ssci.2013.04.001>.
- [162] Khakzad N, Khan F, Amyotte P. Dynamic risk analysis using bow-tie approach. *Reliab Eng Syst Saf* 2012;104:36–44. <https://doi.org/10.1016/j.res.2012.04.003>.
- [163] Ding L, Khan F, Ji J. A novel approach for domino effects modeling and risk analysis based on synergistic effect and accident evidence. *Reliab Eng Syst Saf* 2020;203:107109. <https://doi.org/10.1016/j.res.2020.107109>.
- [164] Hu J, Khan F, Zhang L. Dynamic resilience assessment of the Marine LNG offloading system. *Reliab Eng Syst Saf* 2021;208:107368. <https://doi.org/10.1016/j.res.2020.107368>.
- [165] Laughery KR Jr, Archer S, Corker K. Modeling human performance in complex systems. *Handbook of Industrial Engineering* 2001:2407–44. <https://doi.org/10.1002/9780470172339.ch93>.
- [166] Lebiere C, Gonzalez C, Warwick W. Convergence and constraints revealed in a qualitative model comparison. *J Cognit Eng Decis Mak* 2009;3(2):131–55. <https://doi.org/10.1518/155534309x441880>.
- [167] Anderson JR, Matessa M, Lebiere C. ACT-R: a theory of higher level cognition and its relation to visual attention. *Hum Comput Interact* 1997;12(4):439–62. https://doi.org/10.1207/s15327051hci1204_5.
- [168] Yang K, Tao L, Bai J. Assessment of flight crew errors based on THERP. *Proc Eng* 2014;80:49–58. <https://doi.org/10.1016/j.proeng.2014.09.059>.
- [169] Bowo LP, Furusho M. Human error assessment and reduction technique for marine accident analysis: the case of ship grounding. *Trans Navig* 2018;3(1):1–7. <https://doi.org/10.18949/jintransnavi.3.1.1>.
- [170] Cooper SE, Ramey-Smith AM, Wreathall J, Parry GW. A technique for human error analysis (ATHEANA). U.S. Nuclear Regulatory Commission 1996:1–114. <https://doi.org/10.2172/249298>.
- [171] Bristow DN, Bristow M. Recovery planning for resilience in integrated disaster risk management. In: *Proceedings of the 2017 IEEE international conference on systems, man, and cybernetics (SMC)*; 2017. p. 2643–8. <https://doi.org/10.1109/SMC.2017.8123024>.
- [172] Baidya PM, Sun W. Effective restoration strategies of interdependent power system and communication network. *J Eng* 2017;2017(13):1760–4. <https://doi.org/10.1049/joe.2017.0634>.
- [173] MacKenzie CA, Barker K. Empirical data and regression analysis for estimation of infrastructure resilience with application to electric power outages. *J Infrastruct Syst* 2013;19(1):25–35. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000103](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000103).
- [174] Reilly AC, Samuel A, Guikema SD. Gaming the system": decision making by interdependent critical infrastructure. *Decis Anal* 2015;12(4):155–72. <https://doi.org/10.1287/deca.2015.0318>.
- [175] Sharkey TC, Cavdaroglu B, Nguyen H, Holman J, Mitchell JE, Wallace WA. Interdependent network restoration: on the value of information-sharing. *Eur J Oper Res* 2015;244(1):309–21. <https://doi.org/10.1016/j.ejor.2014.12.051>.
- [176] Nyarko Y, Schotter A. An experimental study of belief learning using elicited beliefs. *Econometrica* 2002;70(3):971–1005. <https://doi.org/10.1111/1468-0262.00316>.
- [177] Ruesch L, Tarakci M, Besiou M, Van Quaquebeke N. Orchestrating coordination among humanitarian organizations. *Prod Oper Manag* 2022;31(5):1977–96. <https://doi.org/10.1111/poms.13660>.
- [178] Tariq H, Pathirage C, Fernando T. Measuring community disaster resilience at local levels: an adaptable resilience framework. *Int J Disaster Risk Reduct* 2021;62:102358. <https://doi.org/10.1016/j.ijdrr.2021.102358>.
- [179] Rak J. Principles of communication networks resilience. In: Rak J, editor. *Resilient routing in communication networks*. Cham: Springer International Publishing; 2015. p. 11–43.
- [180] Li N, Wang F, Magoua JJ, Fang D. Interdependent effects of critical infrastructure systems under different types of disruptions. *Int J Disaster Risk Reduct* 2022;81:103266. <https://doi.org/10.1016/j.ijdrr.2022.103266>.
- [181] Mao Q, Li N. Assessment of the impact of interdependencies on the resilience of networked critical infrastructure systems. *Nat Hazards* 2018;93. <https://doi.org/10.1007/s11069-018-3302-3>.
- [182] Magoua JJ, Wang F, Li N, Fang D. Incorporating the human factor in modeling the operations of interdependent infrastructure systems. *Autom Constr* 2023. Under re-review.